

LIM ZE WEI

Software Engineer | Full-Stack & Backend Systems | Integrations | Cloud & AI

Malaysia | +60 12-279 8096 | limzewei123@gmail.com

PROFESSIONAL SUMMARY

Software Engineer with 3+ years of experience designing, building, integrating, deploying, and supporting production software across backend systems, web applications, Linux infrastructure, networking, and AI. Experienced in delivering end-to-end solutions including REST APIs, third-party integrations, reservation and payment workflows, cloud-controlled gateway systems, automation, and self-hosted LLM services. Comfortable working with existing codebases, investigating production issues, planning implementations, integrating external services, and taking features from development through deployment and support.

CORE SKILLS

Backend & Web: PHP / Laravel, Python, JavaScript, REST APIs, WebSocket, AJAX, HTML/CSS, SQL / MariaDB

AI & Automation: LLM integration, Llama 3, vLLM, Ollama, Redis, local GPU inference, intent/extraction workflows, hybrid rule + LLM systems

Infrastructure & Deployment: Linux (Rocky / Oracle), Docker, Bash, systemd, Apache, SSH, GitHub Actions, server provisioning, deployment, troubleshooting

Networking: DHCP, NAT, iptables/nftables, tc/HTB, Open vSwitch, VLAN, policy routing, SNMP, DNS

Integrations: Meta Messenger, LINE, WeChat, Oracle OHIP, Shiji PMS, PayPal

Engineering: Git, API testing, PHPUnit, system design, debugging, code review, implementation planning, mentoring

PROFESSIONAL EXPERIENCE

Software Engineer | Ntech Systems

2023 - Present

Puchong, Malaysia

- Own design, development, deployment, and production support for hospitality software and gateway products across backend applications, integrations, networking, and Linux infrastructure.
- Led cloud-control modernization of an on-premise hotel internet gateway platform supporting a 100+ hotel install base, using WebSocket-based centralized management for remote configuration, synchronization, and operations.
- Maintain and troubleshoot captive portal, DHCP, NAT/firewall, bandwidth control, VLAN/Open vSwitch, SNMP, routing, storage, and high-availability workflows across production environments.
- Designed and developed a hotel reservation chatbot and administration platform supporting PMS-integrated and local-fallback inventory, pricing overrides, reservation flows, and payment links.
- Integrated Meta Messenger, LINE, WeChat, Oracle OHIP, Shiji PMS, and PayPal through server-to-server APIs and secure message-processing workflows.
- Introduced a hybrid LLM architecture in which the model handles intent detection, information extraction, and response generation while backend services enforce booking rules, session state, PMS calls, and payment logic.
- Deployed and benchmarked self-hosted LLM services using Llama 3, vLLM/Ollama, Redis, and multi-GPU Tesla V100 infrastructure, evaluating latency, throughput, tensor parallelism, and production suitability.
- Contributed to an IPTV platform covering RTSP-to-web delivery workflows, configuration tools, administration, and operational reporting.
- Lead day-to-day engineering work in a small team through planning, implementation guidance, code review, production troubleshooting, task coordination, and mentoring interns.

SELECTED PROJECTS

AI Business OS / Transaction Automation | *Independent Project - In Development*

2026 - Present

- Developing a modular, AI-ready operations platform for freelancers and SMEs covering enquiry-to-payment workflows, customer profiles, quotations, appointments/orders, finance, HR, and IT operations.
- Designed a shared workflow, permission, and data model allowing human users and AI agents to operate through the same underlying business processes rather than maintaining separate AI-only logic.
- Building an integration-first architecture for external CRM, accounting, POS, and automation systems, with planned integrations including Xero, HubSpot, Zoho, and AutoCount.
- Using the project to explore AI-assisted software engineering workflows, modular backend architecture, integrations, automation, and production-oriented system design.

agentCTL | *Private Developer Tooling Project*

2026

WSL / Linux / Windows

- Developing private tooling to install, synchronize, and standardize AI-agent skills, MCP/configuration, and repository instructions across multiple coding-agent environments.
- Built repository synchronization workflows and CI validation using GitHub Actions.
- Focused development and testing on repeatable WSL/Linux and Windows-based AI-assisted coding workflows.

PRODUCT & DOMAIN EXPERIENCE

Hospitality Internet / HSIA: Captive portal, DHCP, NAT, firewalling, bandwidth control, VLAN/OVS, SNMP, routing, high availability, remote cloud control

Reservation & PMS Systems: Multi-channel booking, PMS inventory/rate integration, fallback inventory, payment workflows, Oracle OHIP, Shiji

AI Systems: Self-hosted model serving, intent/extraction pipelines, Redis-backed state, performance benchmarking, backend/LLM responsibility separation

Production Operations: Linux troubleshooting, MariaDB recovery, Apache issues, DHCP corruption, filesystem pressure, backup/restore, server provisioning, networking

INFRASTRUCTURE & PRODUCTION OPERATIONS

- Troubleshoot production incidents across Linux services and infrastructure, including MariaDB/Aria/InnoDB recovery, Apache startup failures, DHCP lease corruption, filesystem pressure, and abnormal log growth.
- Perform server provisioning, disk imaging/restore, storage migration, backup/recovery, and service validation for on-premise gateway deployments.
- Work directly with routing, DNS, firewall/NAT, VLAN, traffic-control, and multi-WAN issues to isolate problems across application, operating-system, and network layers.

EDUCATION

Bachelor of Information Technology (Hons.) - Security Technology

Multimedia University, Malaysia | 2020 - 2023 | CGPA: 3.46

Diploma in Information Technology

Multimedia University, Malaysia | 2018 - 2020 | CGPA: 3.54 | Dean's List (2019), GPA 3.70

LANGUAGES

Bahasa Malaysia | English | Mandarin | Cantonese | Korean